Models with a Single Categorical Predictor (Independent t-test).

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

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Categorical Predictors

- Thus far, we have looked at the relationships between **continuous** predictor variables (X's) and **continuous** outcome variables (Y's).

- Now, we will shift our focus to **categorical** predictors (X's) and **continuous** outcome variables (Y's).
  - Categorical predictors can be "natural" groupings (e.g., a cohort study) or could be experimental groupings (e.g., a randomized controlled trial).

  - Categorical variables can also be "made" by breaking continuous variables into groups. **However**, breaking a continuous variable into a categorical variable is generally a bad practice and has a negative effect on statistical power.

How do we represent a categorical variable?

- **<u>Contrast codes</u>**:
- $\sum \lambda_k = 0$
  - Note this is the sum of λ_k (the code) not the sum of X_i (the data).
  - $\sum \lambda 1_k = 0$
  - $\sum \lambda 2_k = 0$
  - $\sum \lambda 3_k = 1$

Group	X_1	X_2	X_3
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Non-Smoker	-1	-0.5	0
Non-Smoker	-1	-0.5	0
Non-Smoker	-1	-0.5	0
Non-Smoker	-1	-0.5	0

$\sum X_1 = 1$ $\sum X_2 = 0.5$ $\sum X_3 = 5$

How do we represent a categorical variable?

- **<u>Contrast codes</u>**: the levels of our code sum to zero.
  - Equivalent to mean-centering of continuous variables.

- **<u>Treatment/ 'Dummy' codes</u>**: levels of the code do not sum to zero. One group is coded as zero across all levels of the code.
  - Analogous to having an interpretable 0 in your continuous variable.

Group	X_1	X_2	X_3
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Smoker	+1	+0.5	1
Non-Smoker	-1	-0.5	0
Non-Smoker	-1	-0.5	0
Non-Smoker	-1	-0.5	0
Non-Smoker	-1	-0.5	0

Using Contrast Codes in Practice

Student	Group	SAT
1	Course	580
2	Course	560
3	Course	660
4	Course	620
5	Course	600
6	Course	580
7	Course	590
8	Course	640
9	Course	620
10	Course	600
11	No Course	580
12	No Course	530
13	No Course	590
14	No Course	550
15	No Course	610
16	No Course	590
17	No Course	600
18	No Course	530
19	No Course	590
20	No Course	600

- If we want to test the efficacy of an SAT prep course, we would have a model comparison:
  - A: $y_i = \beta_0 + \beta_1(X_{1i}) + \epsilon_i$
  - C: $y_i = \beta_0 + \epsilon_i$

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- If we want to test the efficacy of an SAT prep course, we would have a model comparison:
  - A: $y_i = \beta_0 + \beta_1(X_{1i}) + \epsilon_i$
  - C: $y_i = \beta_0 + 0(X_{1i}) + \epsilon_i$
- Note that in Model C we are hypothesizing there is no effect of an SAT prep course.
  - In Model A, we are allowing for a non-zero effect.

Using Contrast Codes in Practice

Student	Group	SAT	X_1
1	Course	580	1
2	Course	560	1
3	Course	660	1
4	Course	620	1
5	Course	600	1
6	Course	580	1
7	Course	590	1
8	Course	640	1
9	Course	620	1
10	Course	600	1
11	No Course	580	-1
12	No Course	530	-1
13	No Course	590	-1
14	No Course	550	-1
15	No Course	610	-1
16	No Course	590	-1
17	No Course	600	-1
18	No Course	530	-1
19	No Course	590	-1
20	No Course	600	-1

- We can create a contrast code for the Group variable:
  - A: $y_i = \beta_0 + \beta_1(X_{1i}) + \epsilon_i$
- Using a coding scheme of +1 and -1, we get two different estimates for our different groups in Model A:
  - **Course**: $y_i = \beta_0 + \beta_1(\mathbf{1}) + \epsilon_i$
  - **No Course**: $y_i = \beta_0 + \beta_1(\mathbf{-1}) + \epsilon_i$
- And this is a contrast code because:
  - $\sum \lambda_k = 1 + (-1) = 0$

Error and 'Fitting' Model C

Student	Group	SAT	X_1	$\widehat{y_{ci}}$	e_{ci}	e_{ci}^2
1	Course	580	1	591	-11	121
2	Course	560	1	591	-31	961
3	Course	660	1	591	69	4761
4	Course	620	1	591	29	841
5	Course	600	1	591	9	81
6	Course	580	1	591	-11	121
7	Course	590	1	591	-1	1
8	Course	640	1	591	49	2401
9	Course	620	1	591	29	841
10	Course	600	1	591	9	81
11	No Course	580	-1	591	-11	121
12	No Course	530	-1	591	-61	3721
13	No Course	590	-1	591	-1	1
14	No Course	550	-1	591	-41	1681
15	No Course	610	-1	591	19	361
16	No Course	590	-1	591	-1	1
17	No Course	600	-1	591	9	81
18	No Course	530	-1	591	-61	3721
19	No Course	590	-1	591	-1	1
20	No Course	600	-1	591	9	81

- C: $y_i = \beta_0 + \epsilon_i$
- C: $y_i = 591 + e_i$

- $SSE(C) = \sum(y_i - \widehat{y_{ci}})^2 = \sum(e_{ci})^2$
- $SSE(C) = 19{,}980$

Error and 'Fitting' Model A

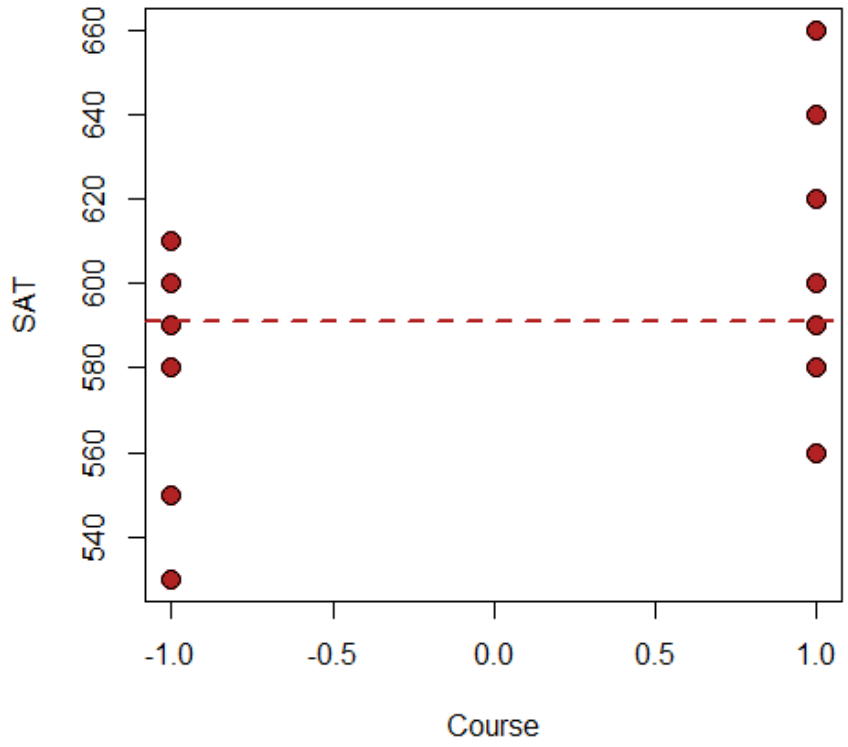
Student	Group	SAT	X_1	$\widehat{y_{ai}}$	e_{ai}	e_{ai}^2
1	Course	580	1	605	-25	625
2	Course	560	1	605	-45	2025
3	Course	660	1	605	55	3025
4	Course	620	1	605	15	225
5	Course	600	1	605	-5	25
6	Course	580	1	605	-25	625
7	Course	590	1	605	-15	225
8	Course	640	1	605	35	1225
9	Course	620	1	605	15	225
10	Course	600	1	605	-5	25
11	No Course	580	-1	577	3	9
12	No Course	530	-1	577	-47	2209
13	No Course	590	-1	577	13	169
14	No Course	550	-1	577	-27	729
15	No Course	610	-1	577	33	1089
16	No Course	590	-1	577	13	169
17	No Course	600	-1	577	23	529
18	No Course	530	-1	577	-47	2209
19	No Course	590	-1	577	13	169
20	No Course	600	-1	577	23	529

- A: $y_i = \beta_0 + \beta_1(X_1)\epsilon_i$
- A: $y_i = 591 + 14(X_1) + e_i$

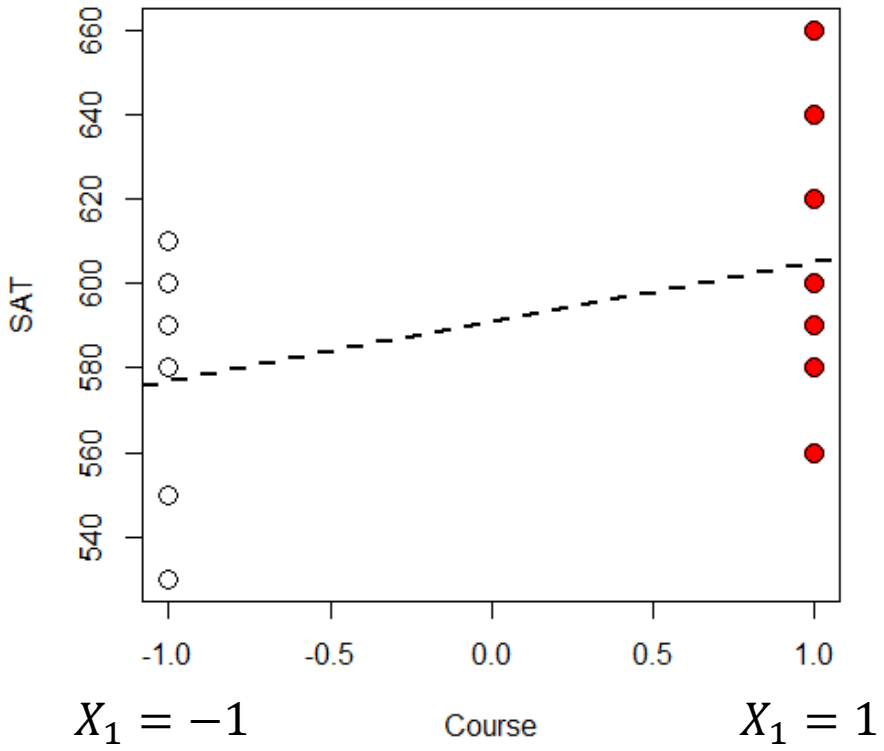
- $SSE(A) = \sum(y_i - \widehat{y_{ai}})^2 = \sum(e_{ai})^2$
- $SSE(A) = 16{,}060$

Comparing the Two Models

Fitted Model C: $y_i = 591 + e_i$
$SSE(C) = 19{,}980$



Fitted Model A: $y_i = 591 + 14(X_1) + e_i$
$SSE(A) = 16{,}060$



ANOVA Table Testing the Null Hypothesis

Source	SSE	Df	MSE	F_obs	Pr(>F_obs)
Regression	3,920.00	1	3,920.00	4.39	0.0505
Error	16,060.00	18	892.22		
Total	19,980.00	19			

Statistical Summary from R

```
lm(formula = SAT ~ Course, data = DATA)

Residuals:
   Min      1Q Median      3Q     Max
   -47     -25      8      17      55

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  591.000      6.679  88.484   <2e-16 ***
Course        14.000      6.679   2.096   0.0505 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 29.87 on 18 degrees of freedom
Multiple R-squared:  0.1962,	Adjusted R-squared:  0.1515
F-statistic: 4.394 on 1 and 18 DF,  p-value: 0.05048
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But how do we get those estimates?

- At the moment, we have let the statistical program calculate the estimate of the intercept and the estimate of the slope for us.
  - These are the best fitting intercepts and slopes in our sample, but they are estimates of the true effects in the population.
  - In the next lecture, we will walk through this process in detail, but these estimates are still based on least-squared error estimation that we have done before.